GS ↔ M2 SerDes link: settings, measurements, open issues

ULX5M-GS ↔ ULX5M-M2 · GateMate CC_SERDES · board-to-board, 8b/10b + PRBS checker · status 17 Oct 2026

REGOČ

PRJ-051 · ULX5M-M2 SerDes/PCIe

1 · SETTINGS

2.5 Gb/s — recipe R1 LINKS

Rate / bits	N1·N2·N3 = 1·5·5, OUTDIV 2, PROFILE 1, TX_NEG 1 · Pi bits `gs/m2_2g5_auto`
CAL0	RX/TX_CALIB_OVR 1, CALIB_VAL 0 (JTAG, after load)
TX_SEL_PRE	0 (POST 5)
DFE / EQA reset	strobe 0x2B after load
TX_AMP	12
RX_EQA_CKP_OFFSET	intended 0 → readback **1**: R1 ran with 1
RX	EQA on · AFE PEAK 24, GAIN 0 · CDR CKP 0x3E

Toolchain DEFAULT

Suite	`oss-cad-suite-20261016`
Versions	yosys 0.70+145 · nextpnr-himbaechel 0.11.1-63 · gmpack v1.13-6 · openFPGALoader v1.1.1
Qualified	CC_SERDES 0/252 fields vs lab bits · CFGRST ✓ · 0 hold violations
R1 baked	gs `79b07795` / m2 `4a94fc99` (4 fields) — not yet loaded

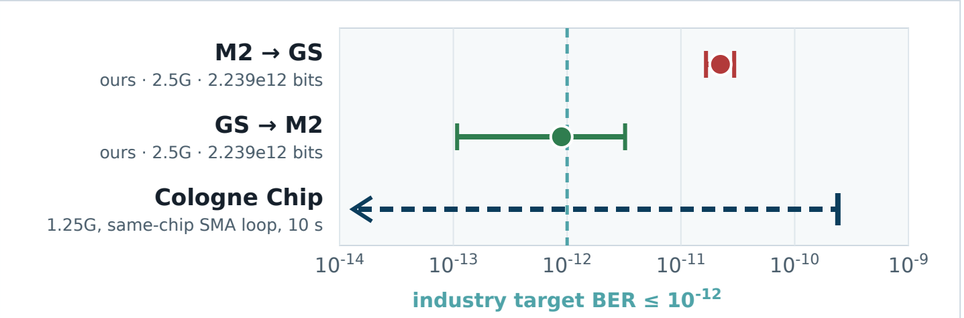
5 Gb/s — P2 NO LOCK

PROFILE 2 bits gs `bec33f07` / m2 `ac8132bd`, SET_ACC 2→0 · gate: ≥ 99 % of 1000 samples PEER

VARIANT	TX_AMP	OTHER M2→GS	OTHER GS→M2
CAL0	31	6.9 % [5.9, 8.1]	8.0 % [6.8, 9.2]
R1	12	10.3 % [9.3, 11.5]	23.9 % [22.4, 25.5]
R1	16	9.4 % [8.4, 10.5]	15.2 % [13.9, 16.5]

OTHER = sample without K28.5 + peer ID · Wilson 95 % CI

2 · 2.5G BER PER DIRECTION vs PUBLISHED DATA



LINK	BIT ERRORS	BER	95 % CI (Poisson)	ERR. WINDOWS
● M2 → GS	50	**2.23e-11**	[1.66e-11, 2.94e-11]	18 / 177
● GS → M2	2	**8.93e-13**	[1.08e-13, 3.23e-12]	2 / 177
● Cologne Chip	0 in 1.25e10	—	upper bound < 2.4e-10	—

Our soak: R1, 1800 s (177 × 10 s windows), 0 sync losses, 17 Oct 00:26. 40 checked bits per word.

What others publish

CHIP / SOURCE	RATE · CHANNEL	PUBLISHED RESULT
Cologne Chip GateMate	2.5G / 5G	no BER, eye or duration (DS1001, UG1001, talks, repos)
Xilinx Artix-7 GTP · WP428	6.25G · 30" FR4	eye scan to 1e-8, extrapolated to 1e-15
Gowin GW5AT · WP1157	to 12G · cables	target 1e-12, PRBS15, "eye collapse"; no duration
CESYS Artix-7 · WP-101	6.25G · HDMI cable	"0E0 errors", eye > 75 %; no duration
Lattice ECP5 · Intel C10GX · Efinix	—	specs at 1e-12 or tools only; no bit counts
Attempts-to-link: published by nobody — our 5/15 has no public reference.		

> **Proof rule:** BER ≤ 1e-12 @ 95 % = 3.0e12 error-free bits = **40 min** on our 2.5G checker (with 2 errors: 6.3e12 bits, 84 min). **GS→M2**: upper bound 2.8e-12, ≈ 54 min more without a new error. **M2→GS**: ≈ 22× above target, no duration can prove it.

3 · OPEN ISSUES

1 — 2.5G link gate passes 5 / 15

L3: 33 % [15 %, 58 %] (Wilson). Soak gate passed on attempt 9: 7 × "Q errors > 8", 1 × m2 RX_CDR_LOCKED = 0. No reset level (L1, L1b, L2tx, L2rx) shown to work.

2 — RX_EQA_CKP_OFFSET is not written

set 0 → readback 1; set 2 → readback 1 (0x04[7:0], both boards). `link_gate.py` does not check readback, so all "R1" data is with OFFSET = 1.

3 — M2 → GS is the weaker direction

50 vs 2 bit errors in 2.239e12 bits; worst 10 s window 5.6e-10. 5 of 7 gate "Q" failures were M2 → GS.

4 — 5G: no CDR lock, 9 / 9 attempts fail

All fail health/PEER before BER (no soak). gs RX_CDR_LOCKED mostly 0; 7–24 % invalid samples. PLL locked, CAP_FT_OF/UF = 0, RX_EQA_LOCKED 1: the 5G eye is not open.

5 — VDD_SER not measured

VDD_SER / VDD_SER_PLL (R105/R106 = 2 Ω) not measured in this lab; supply can be neither confirmed nor ruled out. Next: TP10 / TP8 on both boards while the 5G bit runs.